

You will get two nodes name with the node1.domain<no>.example.com and node2.domain<no>.example.com [ex: node1.domain7.example.com & node2.domain7.example.com]

Read the instructions given as it contain few important values.

- All the users password should be “postroll” unless otherwise specified.
- Node1 IP address should be set to 172.24.7.10, net mask 255.255.255.0, gateway 172.24.7.254 and DNS 172.24.7.254

First set of questions should be completed in the node1 machine. You do not want to worry about the node2 until you complete the node1 machine questions.

Below questions should be performed on node1.domain7.example.com

1. Configure the network with the given values in the instruction section and hostname to the node1.domain.example.com

Answer:

```
hostnamectl set-hostname node1.domain7.example.com ( No need to perform any service restart)
nmcli connection show (Already there is a connection called “System eth0” and the interface is eth0”
nmcli con mod “System eth0” ipv4.address 172.24.7.10/24 ipv4.gateway 172.24.7.254 ipv4.dns
172.24.7.254 ipv4.method manual connect.autoconnect yes
nmcli connection reload
systemctl restart NetworkManager
```

Note: Though you have correctly configure the network interface, it is not possible to log in via SSH due to direct root is disable for SSH. Therefore, until you configure users in the Node1 machine you can not access via SSH.

2. Yum repository is given in the intruction page. Configure the yum repositories

BaseOS url = http://content.example.com/rhel 9/x86_64/dvd/BaseOS
AppStream url = http://content.example.com/rhel9/x86_64/dvd/AppStream

Answer:

```
dnf config-manager --add-repo=http://content.example.com/rhel 9/x86_64/dvd/BaseOS
This will add a repo configuration file in /etc/yum.repos.d/content.example.com_rhel
9/x86_64_dvd_BaseOS.conf
```

Use the vi editor and open the file.

```
vi /etc/yum.repos.d/content.example.com_rhel 9/x86_64_dvd_BaseOS.conf
```

Add the below entry.

```
gpgcheck=0
```

```
dnf config-manager --add-repo=http://content.example.com/rhel9/x86_64/dvd/AppStream
This will add a repo configuration file in /etc/yum.repos.d/content.example.com_rhel
9/x86_64_dvd_AppStream.conf
```

Use the vi editor and open the file.

```
vi /etc/yum.repos.d/content.example.com_rhel 9/x86_64_dvd_AppStream.conf
```

Add the below entry.
gpgcheck=0

Run the **dnf repolist** command to ensure that you have correctly configured.

3. Debug the SELinux issue. A web server running on the port 82 and web server content is hosted under /var/www/html. Troubleshoot the issue and ensure the web site is serving without any issue. Site is serving through the standard http as well. (80/tcp)

Answer:

You can not view the web site as they have not give any URL to connect and check. But if you check the web service status it is failed to start due to non-standard port used in the application (82/tcp). We need to add the 82/tcp port to proper SELinux port label and allow the traffic through the firewall.

```
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
firewall-cmd --add-port=82/tcp
firewall-cmd --add-port=82/tcp --permanent
semanage port -l | grep http (Get the correct http port label to add)
semanage port -a -p tcp -t http_port_t 82
systemctl restart httpd.service
systemctl restart firewall.service
```

4. Create the following users, groups, and group membership:
 - A Group named adminuser. A user “harry” who belongs to adminuser as a secondary group.
 - A user “natasha” who belongs to adminuser as a secondary group. A user “sarah” who does not have access to an interactive shell & who is not a member of sysadm group.
 - “harry”, “natasha”, “sarah” should all have the password of “postroll”.

Answer:

```
groupadd adminuser
useradd -G adminuser harry
useradd -G adminuser natasha
useradd -s /sbin/nologin sarah
passwd <user_name> :- (set password for all users as given)
```

5. Set the cron job for the user “natasha” that should runs daily every 14:23 minute local time executes /bin/echo Haiya.

Answer:

```
su - natasha
crontab -e
23 14 * * * /bin/echo Haiya
```

:wq

To ensure that correctly installed run the below command

crontab -l

Then exit from user natasha with exit command

6. Create a collaborative directory /home/admin with the following characteristics: Group ownership of /home/admin is adminuser. The directory should be readable, writeable, and accessible to member of adminuser, but not to Any other user. Files created in /home/admin automatically have group ownership set to the adminuser group.

Note: (It is understood that root has access to all files and directories on the system)

Answer:

```
[root@node1.com]# mkdir /home/admin
```

```
[root@node1.com]# chgrp adminuser /home/admin
```

```
[root@node1.com]# chmod 2770 /home/admin
```

7. Create user 'alex' with 3456 uid and set the password 'postroll'

Answer:

```
[root@node1.com]# useradd -u 3456 alex
```

```
[root@node1.com]# passwd alex
```

8. Locate all files owned by user aletha and copy it under /root/find

Answer:

```
[root@node1.com]# mkdir /root/find
```

```
[root@node1.com]# find / -user aletha -type f -exec cp -rvp {} /root/natashafiles/ \; 2>/dev/null
```

Or

```
[root@node1.com]# find / -user aletha -type f
```

and user cp command to copy file one by one.

9. Find a string 'asse' from /usr/share/dict/words and put it into /root/lines file.

It is also important that /root/lines files should not contain any spaces and entire word should be in the /root/lines file.

Answer:

```
[root@node1.com]# cat /usr/share/dict/words | grep asse > /root/lines
```

10. Configure autofs to mount users home directory with following manner.

NFS server hosted in 172.24.20.250 /home/remotouser20 directory.

Note the following requirements automount remotouser20 home directory:

remotouser20 is exported on classroom.example.com(172.24.20.250):/home/remotouser20

remotouser20 home directory should be automounted locally beneath /remote as /remote/remotouser20

home directories must be writable by their users.

Answer:

```
dnf install autofs nfs-utils
systemctl enable autofs.service
vim /etc/auto.master.d/remotouser.autofs
    /remote /etc/auto.remotouser

vim /etc/auto.remotouser
    remotouser20 -rw,sync,fstype=nfs4 172.24.20.250:/home/remotouser20
systemctl restart autofs.service
```

Verify NFS auto mount has properly configured

```
ssh remotouser20@node1
touch test
exit
```

11. Create an archive '/root/backup.tar.bz2' of /usr/local directory and compress it with bzip2.

Answer:

```
dnf install bzip2
cd /root
tar -cvf backup.tar /usr/local
bzip2 backup.tar (This will create backup.tar.bz2 archive content)
```

12. Create myscript script to locate /usr/share files size 10 MB or lower and files with the sgid bit enable. The list of files should be copied to /root/script. myscript should be keep under /usr/local/bin/ folder location.

Answer:

```
vim /usr/local/bin/myscript

#!/bin/bash
find /usr/share -type f -size 10M -perm -g=s > /root/script

:wq
[root@node1.com]#chmod +x /usr/local/bin/myscript
[root@node1.com]#myscript
```

13. Configure node1 to get the time synchronization from redhat.domain7.example.com.

Answer:

```
vim /etc/chrony.conf
[Remove the time synchronization configuration lines]
```

Add the below content;

```
server redhat.domain7.example.com iburst
:wq
```

```
systemctl restart chronyd.service
chronyc sources -v
```

14. Create a container image by using the Dockerfile hosted in the below location. The docker image name should be “monitor”.
Dockerfile hosted location: <http://domain.exam.com/rhel9/Containerfile>
Do not modify any content on the Dockerfile. You need to ssh to node1 machine by using user aletha.

Answer:

You can download the Dockerfile to any location. Better to download to user’s home folder.

```
cd /home/aletha
wget http://domain.exam.com/rhel9/Containerfile
dnf install container-tools
podman build -t monitor .
```

15. Create a container by using the above created image. The container name should be “asciipdf”.
It is required to mount /opt/input in the host machine to /opt/incoming in the container, and /opt/output in the host machine to /opt/processed in container. The container should start and stop with the machine bootup. The service name should be container-asciipdf.service

Answer:

First login as root user and perform the below commands

```
[root@node1.com]# mkdir /opt/input
[root@node1.com]# mkdir /opt/output
[root@node1.com]# chown aletha:aletha /opt/input
[root@node1.com]# chown aletha:aletha /opt/output
```

Now log out from the root user and login as aletha user (Use new ssh connection to connect with the node1 machine)

```
[root@node1.com]# loginctl enable-linger aletha
[alth@node1.com]$ podman run -d --name asciipdf -v /opt/input:/opt/incoming:z -v /opt/output:/opt/processed:z <image name>
(Note: You can take the image name with podman images command. (ex: monitor-latest))
[alth@node1.com]$ mkdir -p ~/.config/systemd/user
[alth@node1.com]$ cd ~/.config/systemd/user
[alth@node1.com]$ podman generate system --name asciipdf --files --new
[alth@node1.com]$ systemctl --user daemon-reload
[alth@node1.com]$ systemctl --user enable container-asciipdf
[alth@node1.com]$ systemctl --user start container-asciipdf
```

[alth@node1.com]\$ exit

In order to verify that container is running perfectly. Touch a file in /opt/input and you will get the PDF file in the /opt/output

[root@node1.com]# touch /opt/input/test (check and verify that file is converted or not)

Rest of the questions should required to perform in the node2.domain7.example.com machine

1. Reset the root user password in the node2 machine.

If you do not perform this question you will not be able to continue with the rest of the questions.

Answer:

Reboot the machine and disturb the grub. You will have two Linux kernel lines where you need to modify the rescue kernel not the original kernel lines. This is the new modification happened in the RHEL9 environment.

- **Move the cursor to the rescue kernel entry to boot (the one with the word rescue in its name).**
- **Press e to edit the selected entry.**
- **Move the cursor to the kernel command line (the line that starts with linux).**
- **Delete content until upto the “ro” and Append “rd.break” With that option, the system breaks just before the system hands control from the initramfs to the actual system.**
- **Press Ctrl+x to boot with the changes.**
- **Press Enter to perform maintenance when prompted. (In this stage the kernel will prompt for the password. Just press Enter key and will goes to rd.break mode.)**

To reset the root password from this point, use the following procedure:

```
sh-5.1# mount -o remount,rw /sysroot
```

```
sh-5.1# chroot /sysroot
```

```
sh-5.1# passwd root
```

```
sh-5.1# touch /.autorelabel
```

```
exit
```

```
exit
```

With the two exit machine will go the reboot state and wait for couple minutes to boot the machine.

2. Yum repository is given in the intruction page. Configure the yum repositories
BaseOS url = http://content.example.com/rhel9/x86_64/dvd/BaseOS
AppStream url = http://content.example.com/rhel9/x86_64/dvd/AppStream

Answer:

dnf config-manager --add-repo=http://content.example.com/rhel9/x86_64/dvd/BaseOS This will add a repo configuration file in /etc/yum.repos.d/content.example.com_rhel9/x86_64_dvd_BaseOS.conf

Use the vi editor and open the file.

vi /etc/yum.repos.d/content.example.com_rhel9/x86_64_dvd_BaseOS.conf

Add the below entry.

gpgcheck=0

dnf config-manager --add-repo=http://content.example.com/rhel9/x86_64/dvd/AppStream

This will add a repo configuration file in /etc/yum.repos.d/content.example.com_rhel9/x86_64_dvd_AppStream.conf

Use the vi editor and open the file.

vi /etc/yum.repos.d/content.example.com_rhel9/x86_64_dvd_AppStream.conf

Add the below entry.

gpgcheck=0

Run the **dnf repolist** command to ensure that you have correctly configured.

3. Resize the logical volume “data” to 750M. Make sure in lv volume have some data, data is not effect by resizing

Answer:

It is not required extend the existing Volume Group as the VG has required space to extend the LV. It is required to login to the machine remotely using SSH. But node2 also does not support direct root access via ssh. Therefore, you can create a user with any name and used to access the SSH. Set the user password to “postroll”.

ssh <username>@node2

su – (Gain the root access)

lvdisplay data (Get the respective Volume Group name)

vgdisplay <Volume Group Name> (To ensure that VG have enough space to extend)

lvextend –L 750M /dev/vg_name/data (This volume is formatted using ext4 and need to resize the file system)

resize2fs /dev/vg_name/data

4. Add a swap partition of 512MB and mount it permanently. Do not change any existing SWAP spaces.

Answer:

In the node2 there is second volume /dev/vdb and it has already one partition created with GPT. Use the gdisk command to create the second partition for the swap sapce.

gdisk /dev/vdb

n [hit enter]

Partition Number: 2 [hit enter]

First sector [hit enter]

Last sector +512M [hit enter]

Change the volume type to Linux Swap id is 8300 [hit enter]

w [hit enter]

```
[root@node2.com]# partprobe (This is to update the kernel)
[root@node2.com]# mkswap /dev/vdb2
[root@node2.com]# vim /etc/fstab
(make persistent entry)
/dev/vdb2 swap swap defaults 0 0
[root@node2.com]# mount -a
[root@node2.com]# swapon -a
[root@node2.com]# swapon -s
[root@node2.com]# free -m
```

5. Create a logical Volume with the name of “RHCSA” from the new volume group called “Exam”. The size should be 750M and should mount under the /root/exam with the ext3 partition.

Note: Could not remember the exact question. Either LVM question or just a normal partition with ext3
Answer:

It is required to create new VG and LV. It is required first create a partition with 1GB from the /dev/vdb device.

```
gdisk /dev/vdb
n [hit enter]
Partition Number: 3 [hit enter]
First sector [hit enter]
Last sector +1G [hit enter]
Change the volume type to Linux LVM id is 8e00 [hit enter]
w [hit enter]
[root@node2.com]# pvcreate /dev/vdb3
[root@node2.com]# vgcreate Exam /dev/vdb3
[root@node2.com]# lvcreate -L 750M -n RHCSA Exam
[root@node2.com]# mkfs.ext3 /dev/Exam/RHCSA
[root@node2.com]# mkdir /root/exam
[root@node2.com]# vim /etc/fstab
/dev/Exam/RHCSA /root/exam ext3 defaults 0 0
:wq!
[root@node2.com]# mount -a
[root@node2.com]# df -h
```

6. Configure recommended tuned profile on node2 server.

Answer:

```
[root@node2.com]# dnf install tuned (Most probably service is already installed)
[root@node2.com]# systemctl status tuned; systemctl enable tuned
[root@node2.com]# tuned-adm active
[root@node2.com]# tuned-adm recommend
[root@node2.com]# tuned-adm profile virtual-guest
[root@node2.com]# tuned-adm active
```